Exploring Magnetic Field-Induced Deformations in Fluids To Find Magnetic Susceptibility

Akshat Sharma

University of Minnesota, Minneapolis, MN, USA

(Dated: March 14, 2026)

This paper presents a new method to measure magnetic susceptibility in liquids using the Moses effect. The method relies on the deformation of the liquid surface by a magnetic field that is used to calculate the magnetic susceptibility of the liquid and is detected using the optical lever technique. The optical lever involves using a laser beam incident on the liquid and observing the reflected beam to quantify the changes induced by a magnetic field. Our measurements indicate a magnetic susceptibility to be $-(9.1 \pm 0.9) \times 10^{-6}$ and $(2.4 \pm 0.5) \times 10^{-6}$ for water and 1 Molar solution of Copper Sulphate respectively. Both values are within 1σ of the expected results with a maximum deviation of 10%.

I. INTRODUCTION

The magnetic susceptibility of a material is an extrinsic property that tells us about its degree of sensitivity to magnetic fields. Mathematically, Magnetic Susceptibility (χ) is defined as

$$\chi = \frac{\vec{M}}{\vec{H}}, \tag{1}$$

where $\vec{M}$ is the magnetization in the material (dipole moment per unit volume) and $\vec{H}$ is the auxiliary field (magnetic field for a given unit of permittivity of the surrounding for linear materials) [1].

Materials with a positive susceptibility tend to be attracted to the magnetic field and become magnetized in the same direction as the applied field. These materials are known as paramagnetic. The magnetic field repels materials with a negative susceptibility and tends to become magnetized in the opposite direction of the applied field. These materials are known as diamagnetic. Practical applications for magnetic susceptibility include magnetic resonance imaging (MRI) [2], Spin-tronics [3] and energy conversion using soft magnet materials [4]. Traditionally, measuring magnetic susceptibility, especially in feebly paramagnetic or diamagnetic liquids is a time and resource expending process [5]. Standard techniques like SQUID (superconducting quantum interference device) magnetometers operate on the principle of measuring extremely small changes in magnetic flux through a superconducting loop, utilizing quantum interference effects to achieve high sensitivity and detect the magnetic susceptibility of materials. The process involves a complex apparatus, and specialized equipment that demands significant investments in time, financial resources, and specialized infrastructure [6].

Alternatively, Magnetic susceptibility effects can be measured using High-resolution nuclear magnetic resonance (NMR) by detecting changes in chemical shifts and relaxation times of the material. This technique utilizes concepts like spatial encoding and quantitative analysis to gather information about the magnetic properties of materials.[7] However, a high-resolution NMR of liquids poses challenges for resolution and accurate chemical shift assignment, with greater complexity in heterogeneous samples [8].

This paper aims to present a novel comparative technique that exploits the Moses effect for measuring the magnetic susceptibility of liquids materials. The Moses effect, marked by a relatively weak but noticeable deformation of the surface of diamagnetic liquids under a magnetic field, provides us a way to measure the magnetic susceptibility of liquids, effectively addressing challenges associated with conventional measurement techniques [9]. Despite being a relatively weak phenomenon, with magnetic fields typically ranging from 0.1 to 1T, it results in the formation of a near-surface 'well' with micrometer-scale depth, providing a subtle yet effective indicator of magnetic susceptibility [5]. The proposed approach not only capitalizes on the experimental manifestation of the Moses effect but also integrates optical techniques for precise measurements, offering a more accurate and efficient alternative to conventional methodologies.

II. THEORETICAL FOUNDATION

A. The Moses Effect

The Moses Effect refers to the deformation of a liquid's surface in the presence of a magnetic field, particularly observed in diamagnetic fluids like water. As observed in [9], the energy required to create the phenomenon is proportional to the magnetic energy density of the liquid. Hence, the theoretical foundation for understanding this phenomenon involves a balance between gravitational and magnetic energy densities.

The auxiliary field ($\vec{H}$) can be mathematically described as

$$\vec{H} = \frac{\vec{B}}{\mu_m}, \tag{2}$$

where B is the magnetic field and μ_m is the permeability of the material. The permeability of the material can be expressed in terms of relative permeability (k_m) i.e. $\mu_m = k_m \mu_0$. Similarly, the relative permeability of the material can be expressed in terms of its magnetic susceptibility (χ) using $k_m = 1 + \chi$. So, we get [1]

$$\vec{H} = \frac{\vec{B}}{(1+\chi)\mu_0}. \tag{3}$$

For paramagnetic and diamagnetic materials the relative permeability is very close to 1 and so, the magnetic susceptibility very close to zero. So, it is fair to assume that $\chi << 1$ [10]. So, Eq. (3) becomes

$$\vec{B} \approx \mu_0 \vec{H}. \tag{4}$$

Now, the magnetic energy density (U_m) is defined as [1]

$$U_m \equiv -\vec{B} \cdot \vec{M}. \tag{5}$$

As the magnetic field passes through the liquid, it starts to magnetize the dipoles in the liquids, changing its energy continuously. So, let us say, that for a small change in magnetization ($\mathrm{d}M$), the amount of change in energy is ($\mathrm{d}U_m$) then,

$$\mathrm{d}U_m = -\vec{B} \cdot \mathrm{d}\vec{M}. \tag{6}$$

From Eq. (1), we can claim that $\mathrm{d}\vec{M} = \chi \mathrm{d}\vec{H}^*$. And from Eq. (4), we have $\vec{B} \approx \mu_0 \vec{H}^*$, where H^* is a variable replacement of the auxiliary field H. So, we get

$$U_m = -\int_0^H \mu_0 \chi \vec{H}^* \mathrm{d}\vec{H}^* \tag{7}$$

$$= -\frac{1}{2}\mu_o \chi H^2. \tag{8}$$

To maintain energy equilibrium and avoid the generation of forces due to energy gradients, the negative magnetic energy at the liquid surface (as given by Eq. (8)) must be counterbalanced by an increment in the gravitational energy. The Gravitational potential energy can be accounted for via:

$$u_g = mgh, \tag{9}$$

where m is the mass of the mass of the liquid, g is the gravitational acceleration on Earth and h is the height from the Zero potential point (in this case, the surface level of the liquid away from the magnetic field). Since it is a liquid with volume (V), we can find the gravitational energy density i.e.

$$U_g = \frac{u}{V} \tag{10}$$

$$= \frac{mgh}{V} \tag{11}$$

$$= \rho g h, \tag{12}$$

where ρ represents the density of the liquid. Since the liquid is in static equilibrium, the magnetic energy density is in equilibrium with the gravitational energy density and we can conclude that

$$\Delta U_m + \Delta U_g = 0. \tag{13}$$

We can use Eq. (8) and Eq. (12) in Eq. (13) to find

$$-\frac{1}{2}\mu_0 \chi H^2 + \rho g \Delta h = 0; \tag{14}$$

$$\chi = \frac{2g\rho(\Delta h)}{\mu_0 H^2}. \tag{15}$$

This equilibrium results in the formation of a depression or a peak in the liquid's surface when the susceptibility is negative or positive respectively [11]. However, due to the proportionality of the Moses effect to the square of the magnetic field $\vec{B}$, the deformation Δh becomes even less significant at lower magnetic fields. The Moses effect is non-visible with a permanent magnet with a strength (magnetic field) $< 0.1T$[12].

B. The Optical Lever Technique

The optical lever technique is a method used to analyze the deformation of a liquid's surface,

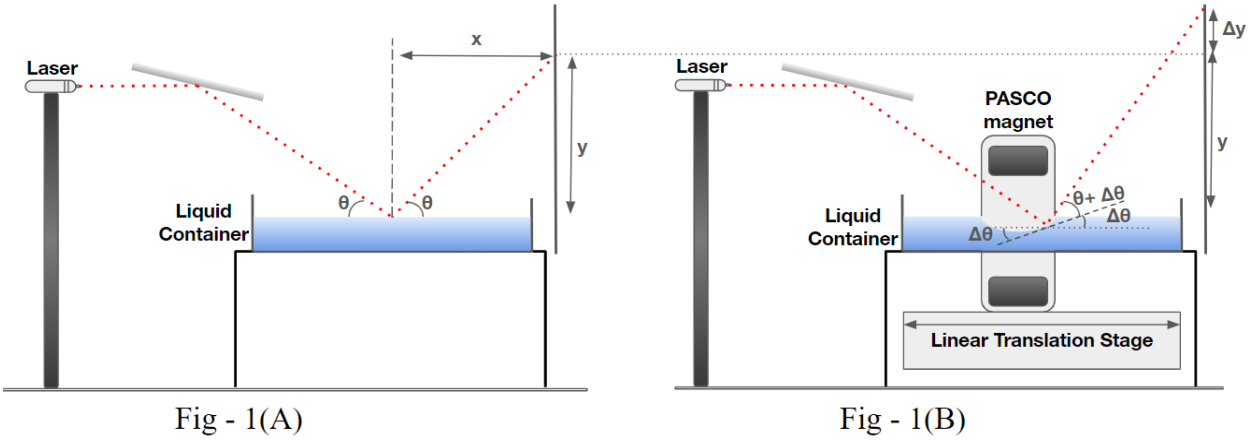

Fig - 1(A) Fig - 1(B)

FIG. 1: The diagram represents the setup of the experiment. Fig - 1(A) shows the calibration of the experiment, where a light beam reflects off the liquid surface and goes towards and hits a wall x distance away from the point of reflection and y distance away from the liquid's horizontal level. The angle produced by the incident light falling on a flat liquid surface is θ. In Fig - 1(B), the liquid is under the influence of a PASCO variable range magnet. The angle made by the incident light on the curved surface is $\theta + \Delta\theta$. The Moses effect shown in the graphic is not drawn according to scale

particularly in the context of the Moses Effect experiment. This technique involves the use of a laser beam and observation of the change in the angle of reflection of the beam to quantify the changes induced by a magnetic field. [11]

As shown in Fig - 1(A), a laser beam is directed to hit the water surface at approximately the center of the dish and reflects off creating an angle θ from the liquid surface. When the water surface is deformed by the magnetic field, the slope of the water's surface changes, leading to an alteration in the angle of incidence of the laser beam, as seen in Fig - 1(B). This change in incident angle ($\Delta\theta$) results in a modification of the reflected beam by an angle of $2\Delta\theta$. The altered reflection angle causes a shift in the position of the reflected laser spot from its initial spot, denoted as Δy in Fig - 1(B) [11].

Using trigonometry on Fig - 1(A), we can claim the following:

$$\tan(\theta) = \frac{y}{x} \tag{16}$$

Now, as we can see in Fig - 1(B), an increment of reflected angle by $2\Delta\theta$ caused the displacement of the laser beam by Δy. Using trigonometry, we can show that,

$$\tan(\theta + 2\Delta\theta) = \frac{y + \Delta y}{x} \tag{17}$$

Combining Eq. (16) and Eq. (17), we can find the value for $\Delta\theta$ is

$$\Delta\theta = \frac{1}{2}\left(\arctan\left(\frac{y + \Delta y}{x}\right) - \arctan\left(\frac{y}{x}\right)\right). \tag{18}$$

As shown in Fig - 1(B), the slope (S) of the liquid's surface relative to the horizontal is given by $\tan(\Delta\theta)$. Experimental data, particularly the

peak/valley in the liquid, is obtained by varying the position of the magnet itself through incremental shifts using a stepper motor over a specified distance [11].

Since measurements are relative, the zero for the magnet position is arbitrary. For consistency, we will be defining zero at the left end of the linear transition stage, as shown in Fig - 1(B)

III. EXPERIMENTAL PROCEDURE

To create the setup shown in Fig - 1(A), we used a plastic petri dish with an 8.5 cm diameter filled with the liquid. A laser, attached to a ring stand, is horizontally pointing towards a mirror, angled at about 80 degrees from the vertical, which reflects the beam towards the liquid surface. The entire setup is positioned at some distance from the wall for observing the reflected laser beam. To calibrate the setup, the laser is turned on before putting the magnet. Calibration of the setup includes measuring the horizontal distance from the point of reflection to the wall (x) and the vertical distance from the top of the liquid level to the point of intersection of the laser and the wall (y) as shown in Fig. - 1(A). [11].

A. placement of magnet

A NdFeB magnet variable range magnet, positioned beneath the dish, is controlled using the stepper motor, allowing it to move from one side of the petri dish to another. The magnet needs to be placed in such a way that it can move through the diameter of the petri dish without blocking the beam of light.

Additionally, for a precise experiment, the center of the magnet must pass through the point of reflection. To ensure that, we used a small iron flake and put it on the liquid surface. The flake was attracted by the magnet and followed its center. This gave us a projection of the center of the magnet on the liquid surface. Then, the magnet was adjusted to make sure that the laser beam intersected with the path of that flake. This ensures that the magnetic field is strongest at the point of reflection when the magnet passes through it.

B. data collection

This experimental procedure involves the gentle movement of the NdFeB magnet using a stepper motor while keeping the liquid still. This ensures minimal disruption and ripples on the water's surface. The laser beam, fixed at a consistent angle, remains directed towards the center of the dish.

To gather data, the magnet's position is altered with respect to the liquid and the laser in incremental steps using the stepper motor. With every increment, the change in corresponding measurements of the reflected laser beam's position on the wall (Δy) is recorded until the

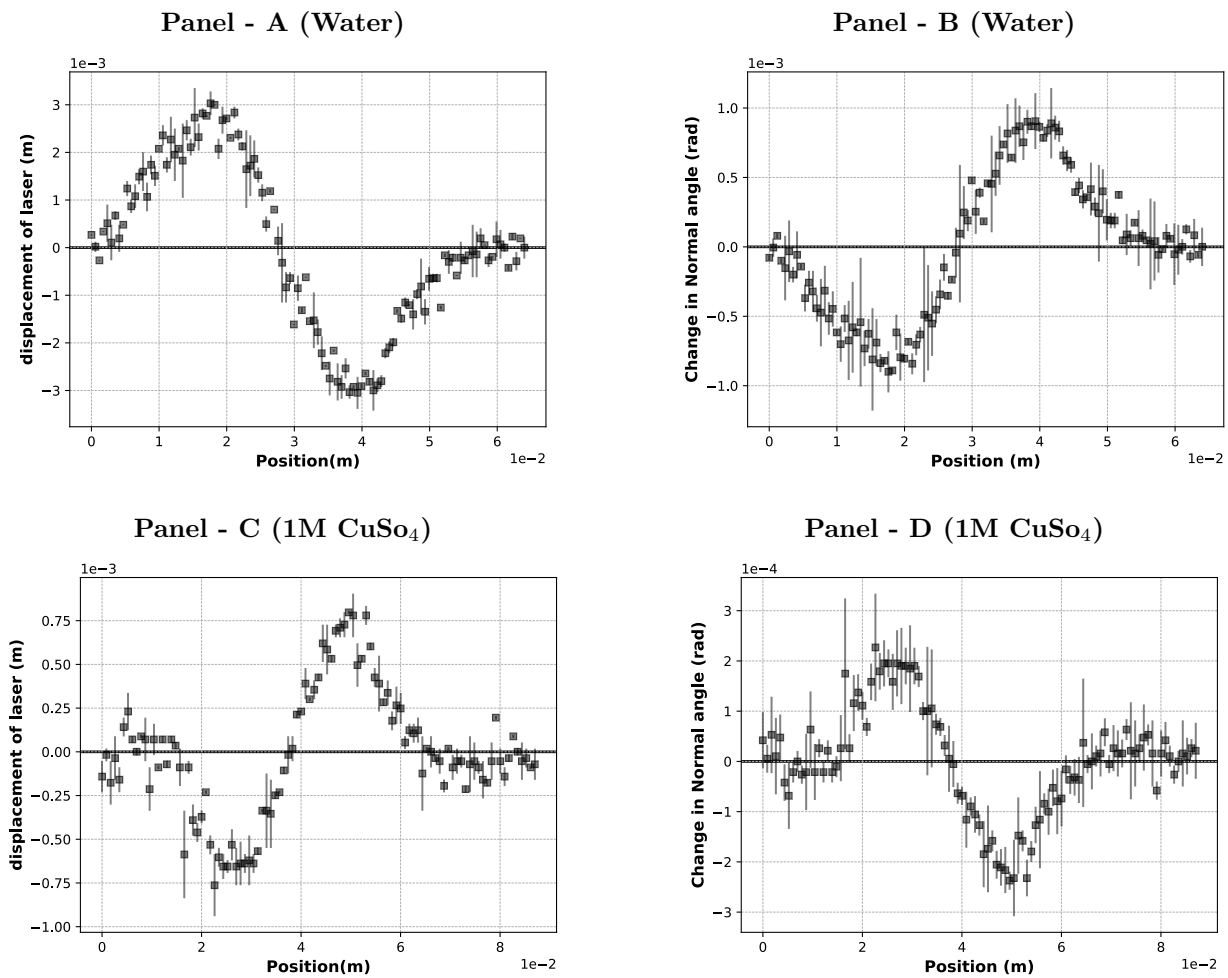

FIG. 2: Panel - A and C represent the displacement of the laser from its original position versus the position of the magnet for water and 1M $CuSo_4$ respectively. Panel - B and D show the angle of the water surface (slope) versus the position of the magnet for water and 1M $CuSo_4$ respectively. For water, the slope starts as negative and then becomes positive, indicating the water surface has a dip. For $CuSo_4$, the slope goes from negative to positive, indicating a peak in the liquid surface.

magnet is past the point of reflection on the petri dish. [11].

ANALYSIS AND RESULTS

For this experiment, we use water, as a diamagnetic material, and 1M Copper Sulphate solution, as a paramagnetic material. The data analysis involves employing trigonometric relations to connect the change in the reflected beam angle ($\Delta\theta$) to the change in height (Δy). The slope of the water's surface is calculated from the measured dimple positions, allowing for the quantitative analysis of the water surface depression induced by the magnetic field

Fig - 2 (A and C) shows us the distance shifted by the laser beam for every change in the position of the magnet. The data shows

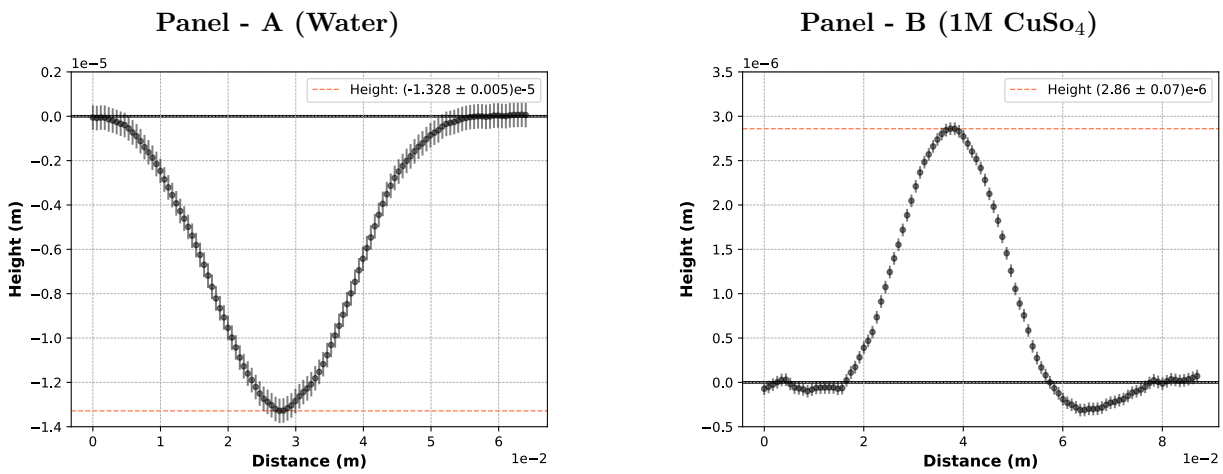

FIG. 3: Panel - A and B represent the displacement of the water level from its original position versus the position of the magnet for water and 1M $CuSo_4$ respectively. So, the maximum displacement in the water surface is $-(1.328 \pm 0.005) \times 10^{-5} m$ and for $CuSo_4$ is $(2.85 \pm 0.07) \times 10^{-6} m$. The displayed error bars in Panel - A are scaled up by a factor of 10, because of their excessively small size.

that the position follows a pattern comparable to the sinusoidal pattern. Uncertainties in the measurements were estimated by conducting multiple measurements of deflection versus position and finding their standard deviation. From this, we can use Eq. (18) to deduce the angle tangential to the water plane.

Fig - 2 (B and D) shows the outcome of the measurement of the angle of reflection of water under the influence of a magnetic field, with respect to the position of the magnet. For our setup, the maximum magnetic field reached $0.19 \pm 0.01T$ for water and $0.17 \pm 0.01T$ for $CuSo_4$.

To compare the surface profile with the measured magnetic fields, we need to derive the displacement of water as a function of position. Under the small angle approximation, the change in the water surface angle can be considered as the slope of the water, enabling us to determine the displacement through Riemann summing. We utilized the trapezoidal rule for Riemann summing, summing both left to right and right to left, and averaging the results. The error was calculated to be the standard deviation of the average from one of the Riemann summations.

As seen in Fig - 3, the Riemann sum provides us with an analogous model of the water surface under the influence of the magnetic field. From here, we can notice that the maximum displacement in the water surface is $-(1.328 \pm 0.005) \times 10^{-5} m$ and for 1M $CuSo_4$ is $(2.86 \pm 0.07) \times 10^{-6} m$.

We can use Eq. (15) to find the magnetic susceptibility of the liquid. This gives us a magnetic susceptibility of $-(9.1 \pm 0.9) \times 10^{-6}$ for water and $(2.4 \pm 0.5) \times 10^{-6}$ for 1M $CuSo_4$ solution.

Considering that the expected magnetic sus-

ceptibility is 9.04×10^{-6} and 2.71×10^{-6} for water and 1M $CuSo_4$ solution respectively. [11] For this dataset, the measured value is approximately 0.6% higher and 10% lower for each liquid, primarily due to a combination of magnetic field error, neglecting surface tension energy, impurities in liquid, and stability of the setup.

IV. CONCLUSION

Applying a combination of the Moses effect and the optical lever technique, we have determined that the magnetic susceptibility is $-(9.1\pm0.9)\times10^{-6}$ for water and $(2.4\pm0.5)\times10^{-6}$ for 1M $CuSo_4$ solution. The change in sign implies that water is diamagnetic, while $CuSo_4$ is paramagnetic. Both results are within 1σ of the expected outcomes. The discrepancy between the expected and measured magnetic susceptibility can be attributed to various factors including errors in the magnetic field measurement, neglecting the influence of surface tension energy, the presence of impurities in the liquid, and potential instability in the setup during the experiment.

The results for both water and $CuSo_4$ are both within 10% of the actual results, which was the expected outcome. [11] Nonetheless, the results for water are much more accurate in comparison to $CuSo_4$. Further research is required with other types of liquids, to find out potential reasons for such discrepancy in the accuracy of the results.

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